$E_2 \to H_a T_b$

$P(E_2) = P(X = 1 \text{ but } H_2) = \underline{0.36}$

(c) $\to$ F (event space) = { $H_a H_b$, $H_a T_b$, $T_a H_b$, ~~$T_b H_a$~~, $T_a T_b$ }

$P(H_a H_b) = 0.24$

$P(H_a T_b) = 0.36$

$P(T_a H_b) = 0.16$

~~$P(T_b H_a)$~~ =

$P(T_a T_b) = 0.24$

$P(F) = 0.24 + 0.24 + 0.36 + 0.16 = 1$

Q 2.2

$\Omega = \{x_1, x_2, x_3, x_4, x_5\}$ $x_i \in R$

$1 \le i \le 5$

$\omega = \{5.76, 4.56, 2.33, 6.32, 5.4\}$

ω is one of them. one of element Ω

~~$X = \{0, 1\}$~~

~~failure ($x_i \not\le 5$)~~

~~success ($x_i \le 5$)~~

$\Omega_x = \{0, 1, 2, 3, 4, 5\}$

$\downarrow$

No of success

describe X as $\omega = \{x_1, x_2, x_3, x_4, x_5\}$

$$I(x_i) = \begin{cases} 1 & x_i \le 5 \text{ success} \\ 0 & x_i > 5 \text{ failure} \end{cases}$$

always define X as function

$$X(\omega) = \sum_{i=1}^{5} I(x_i)$$

X for given data $= 0 + 1 + 1 + 0 + 0$
$= 2$

Sample space for X:

$$\Omega_X = \{0, 1, 2, 3, 4, 5\}$$

(b) $P(\text{success}) = p$

$X \sim$ as binomial random variable

$$P(X = k) = {}^5C_k \, p^k (1-p)^{5-k}, \quad k = 0, 1, 2, 3, 4, 5$$

probability distribution Table.

k	$P(X=k)$
0	${}^5C_0 (1-p)^5$
1	${}^5C_1 \, p(1-p)^4$
2	${}^5C_2 \, p^2(1-p)^3$
3	${}^5C_3 \, p^3(1-p)^2$
4	${}^5C_4 \, p^4(1-p)$
5	${}^5C_5 \, p^5$

Assingment 2

Ques.

1.1 → already solved in Assignment 1.1

1.2 →

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

a) $$f(0) = \frac{1}{\sigma 2\pi} e^{-1/2} \approx 0.242$$

b) $$f(1) = \frac{1}{\sqrt{2\pi}} e^{-1/2} \approx 0.242$$

c)

$$P(x_1 \le X \le x_2) = 0.3$$
$$P(x_1 \le X \le x_3) = 0.45$$

Since

$$P(x_1 \le X \le x_3) = P(x_1 \le X \le x_2) + P(x_2 \le X \le x_3)$$

$$P(x_2 \le X \le x_3) = 0.45 - 0.3 = 0.15$$

$$P(x_2 \le X \le x_3) = 0.15$$

(1.3) →

$$f(x) = \frac{(\alpha+\beta-1)!}{(\alpha-1)!\,(\beta-1)!} x^{\alpha-1}(1-x)^{\beta-1}$$

$$g(x) = x^{\alpha-1}(1-x)^{\beta-1}$$

$$g'(x) = (\alpha-1)x^{\alpha-2}(1-x)^{\beta-1} - (\beta-1)x^{\alpha-1}(1-x)^{\beta-2}$$
$$x^{\alpha-2}(1-x)^{\beta-2}\left[(\alpha-1)(1-x) - (\beta-1)x\right]$$

$g'(x) = 0$ $$x = \frac{\alpha-1}{\alpha+\beta-2}$$

Mode of $X = \frac{\alpha-1}{\alpha+\beta-2}$ for $\alpha > 1$
$\beta > 1$

(1.4) $f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$

compare it with pdf $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$

a) $\mu = 0$
$\sigma = 1 \Rightarrow \sigma^2 = 1$ Mean $= 0$
Variance $= 1$
$X \sim N(0,1)$.

b) probability P of obtaining $X = 0$

$f(0) = \frac{1}{\sqrt{2\pi}} e^{-0^2/2} = \frac{1}{\sqrt{2\pi}}$

$f(0) = 0.399$

c) $P(\cancel{X} < 0) = 0.5$

d) $P(X=0) = 0$ (for a continous random variable : for any exact value x

(2.1) $\rightarrow$ $f(k,\lambda) = \frac{\lambda^k e^{-\lambda}}{k!}$ $\lambda_3 < \lambda_1 < \lambda_2 \rightarrow$ German
Czech english

$f(0,\lambda) = \frac{\lambda^0 e^{-\lambda}}{0!} = e^{-\lambda}$

Since $\lambda_3 < \lambda_1 < \lambda_2$

$e^{-\lambda_3} > e^{-\lambda_1} > e^{-\lambda_2}$

Czech has highest no of words pair with distance 0

2.2 $f(1,\lambda) = \lambda e^{-\lambda}$

$\lambda_1 = 0.7, \approx 10^5$ (total)

$\lambda_2 = 0.8,$ total $= 10^7$ (German)

$\lambda_3 = 0.6,$ total $= 10^8$ (czech)

english $f(1) = 0.7e^{-0.7} \approx 3.48\times10^4$

German $f(1) = 0.8\, e^{-0.8} \approx 3.6\times10^6$

Czech $f(1) = 3.29\times10^7 = 0.6\,e^{-0.6}$

2.3 $f(k) = \dfrac{2^k e^{-2}}{k!} \Rightarrow \lambda = 2$

natural languages: $\lambda_1, \lambda_2, \lambda_3 < 1$

natural language show a strong bias toward short-distance dependencies while random structures produce more long-distance relations due to lack of constraints.

Quesn 3

a) likelihood fun for fixed $x = 220$

$$L(\mu|x = 220) = \frac{1}{220\sqrt{2\pi}} e^{-\frac{(\log 220 - \mu)^2}{2}}$$

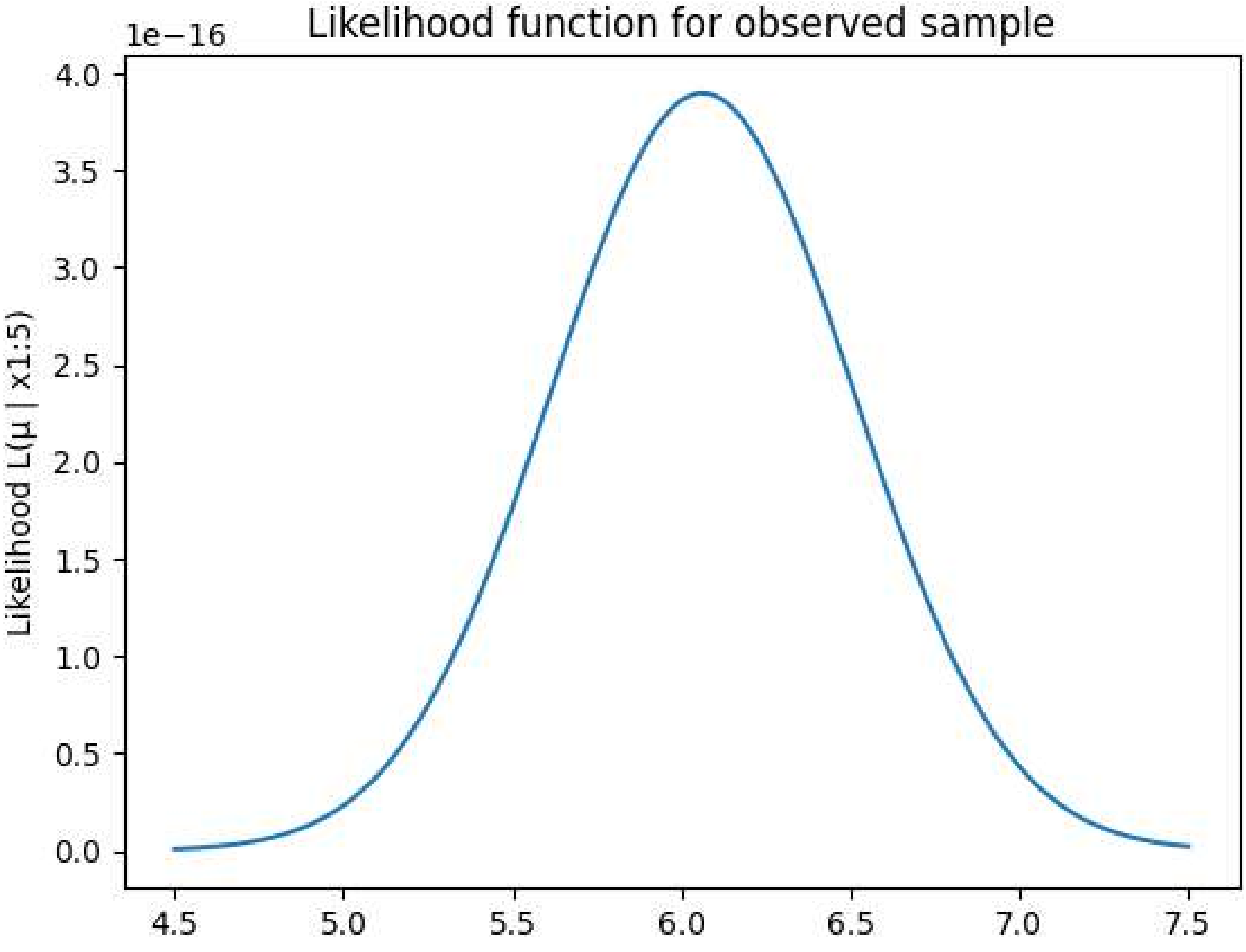
$$\mu = \log(220) \approx 5.4$$

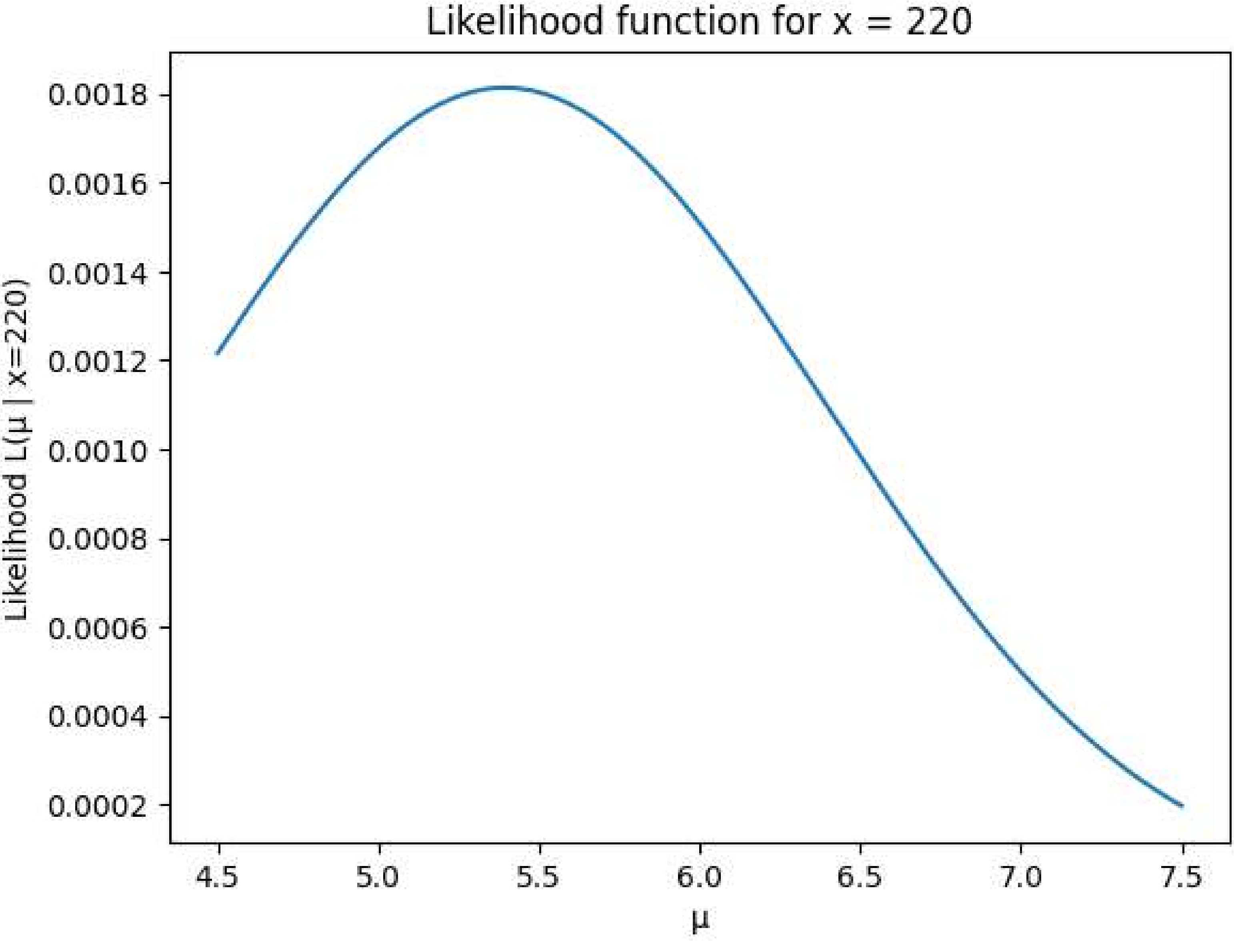
$$\log(220) = 5.39$$

b) $x = [303.25,\ 943,\ 220, 560, 880]$

$$L(\mu|x_{1:5}) = \prod_{i=1}^{5} \frac{1}{x_i\sqrt{2\pi}}\, e^{-\frac{(\log x_i - \mu)^2}{2}}$$

$$= \frac{1}{\prod_i x_i (\sqrt{2\pi})^5}\, e^{-\frac{1}{2}\sum_{i=1}^{5}(\log x_i - \mu)^2}$$


Likelihood function for observed sample
1e−16
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Likelihood L(μ | x1:5)
4.5
5.0
5.5
6.0
6.5
7.0
7.5
μ



Likelihood function for x = 220
Likelihood L(μ | x=220)
0.0018
0.0016
0.0014
0.0012
0.0010
0.0008
0.0006
0.0004
0.0002
4.5
5.0
5.5
6.0
6.5
7.0
7.5
μ

with more data points,
the likelihood curve become
sharper & more peaked

(e) Value of μ that maximizes the likelihood

$$l(\mu) = -\frac{1}{2} \sum_{i=1}^{5} (\log x_i - \mu)^2 + \text{constant}$$

$$\mu = \frac{1}{5} \sum_{i=1}^{5} \log x_i$$

μ_{MLE} = mean of $\log(x_i)$

numerically $\mu_{MLE} \approx 6.06$.

4.1 $\mu = 500$
$\sigma = 100$

a). $P(X \leq 700)$ $z = \frac{700-500}{100} = 2$

$P(X \leq 700) \approx 0.9772$

b) $P(X \geq 900)$ $z = \frac{900-500}{100} = 4$

$P(X \geq 900) \approx 0.00003$

c) $P(200 \leq X \leq 800)$

$z_{800} = 3$, $z_{200} = -3$

$P(200 \leq X \leq 800) \approx 0.9973$

[1]
✓ 0s

```r
pnorm(700, mean = 500, sd = 100)
```

```
0.977249868051821
```

[2]
✓ 0s

```r
1 - pnorm(900, mean = 500, sd = 100)
```

```
3.167124183312e-05
```

[3]
✓ 0s

```r
pnorm(800, mean = 500, sd = 100) - pnorm(200, mean = 500, sd = 100)
```

```
0.99730020393674
```

[4]
✓ 0s

```r
dnorm(x = 550, mean = 650, sd = 125)
```

```
0.00231753242209186
```

[5]
✓ 0s

```r
set.seed(123)     # optional but good practice
xsim <- rnorm(10000, mean = 300, sd = 200)
```

[6]
✓ 0s

```r
mean(xsim)
sd(xsim)
min(xsim)
max(xsim)
```

```
299.525659631332
199.727326925402
-469.064033892007
1069.55354116598
```

7]
0s

```r
hist(xsim, breaks = 50, col = "lightblue",
     main = "Histogram of Simulated Values",
     xlab = "xsim values")
```